AI OPEN
DATA CENTER:
Power Distribution



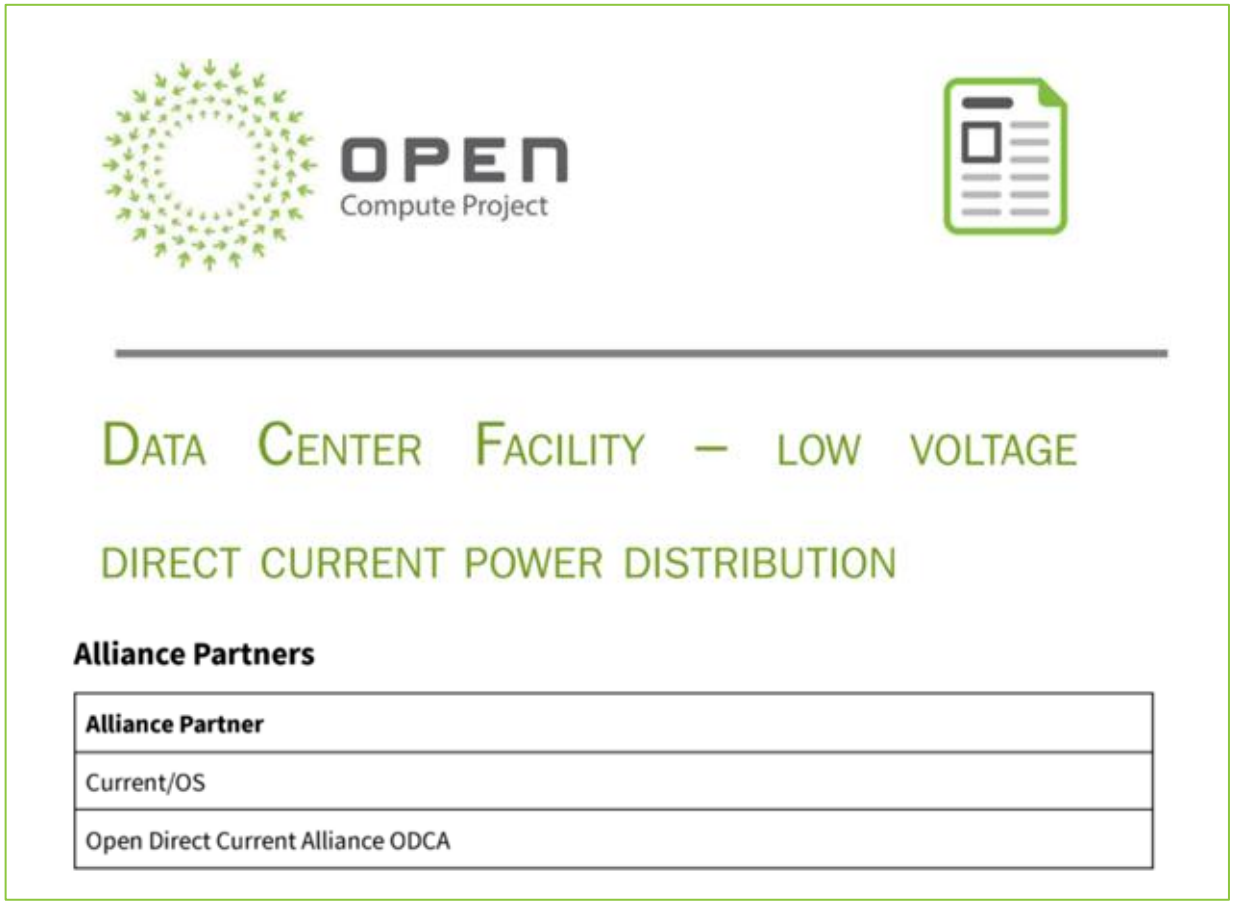
LVDC Architecture Fundamentals:
Current/OS Zones, ODCA Sectors, and IEC Circuits

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OCP PLATINUM® Member

LVDC Power Distribution White Paper



OPEN Compute Project

DATA CENTER FACILITY – LOW VOLTAGE

DIRECT CURRENT POWER DISTRIBUTION

Alliance Partners

Alliance Partner
Current/OS
Open Direct Current Alliance ODCA

Current/OS | ODCA direct current by zvei

Key Contributors:

- Current/OS
- Open DC Alliance
- Data center industry

Updated v1.1
(Sep. 2026)

Direct Current Expertise

CurrentDS | ODCA direct current by zvei



Combined: 210+

- Industry: 150+
- Research/Ed.: 40+
- Certification: ~10
- Trade Groups: ~10

Direct Current projects since 2016

DC Zones: Overview

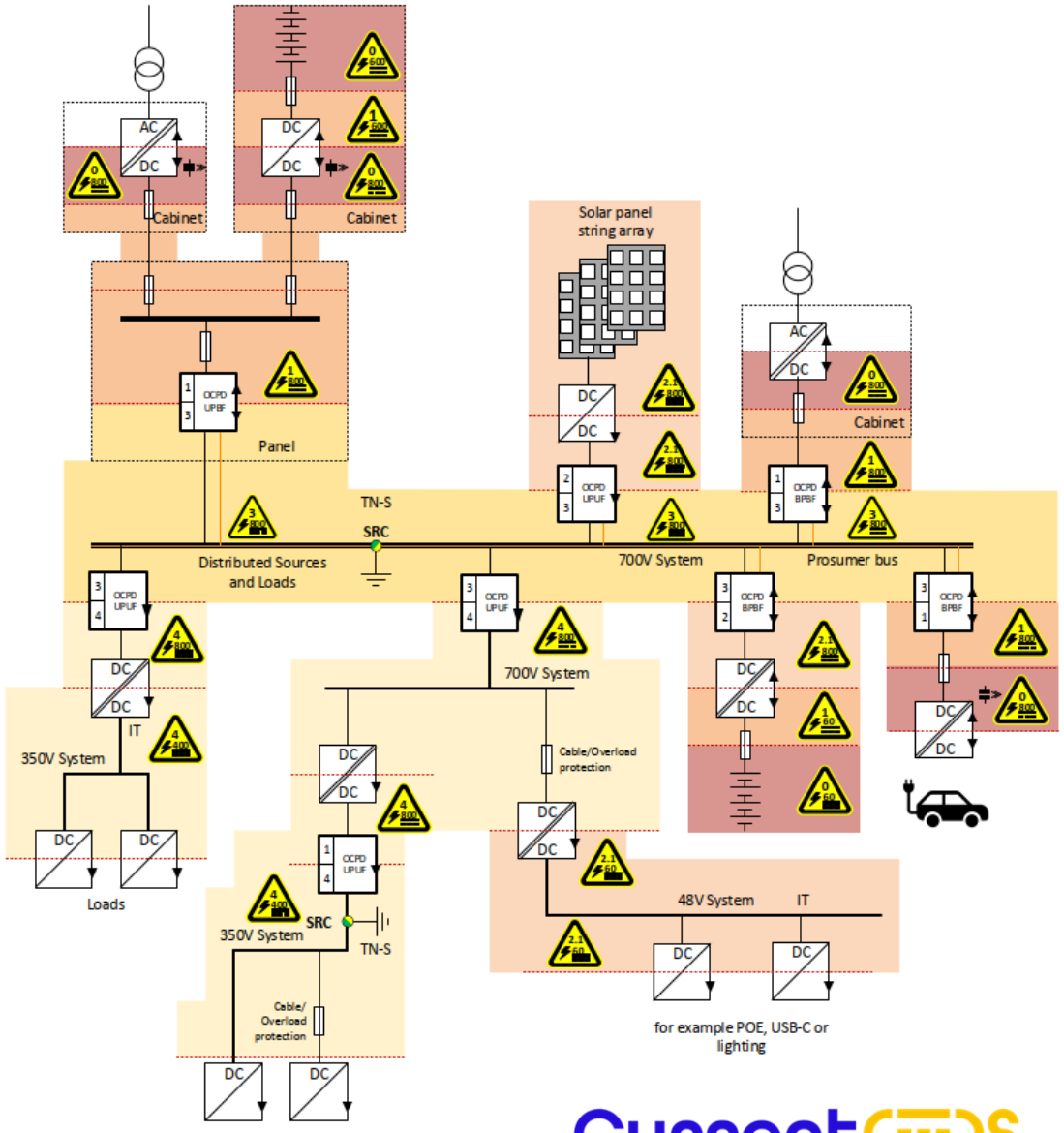
- **DC Zone 0 –** voltage source(s)
- **DC Zone 1 –** voltage source(s) after protection
- **DC Zone 2.x –** current limited source(s), native or by fast protection
- **DC Zone 3 –** multiple distributed sources with ultrafast protection
- **DC Zone 4 –** single source with ultrafast protection

↑ DC zones for **↓ Risk** and **↑ Interoperability**

Low fault energy in DC **Zones 3 and 4**

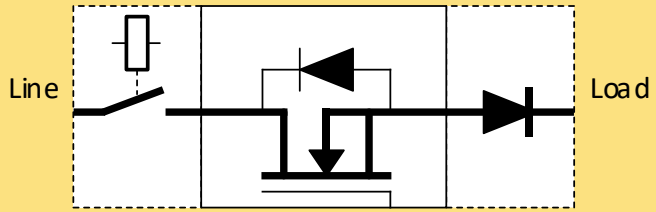
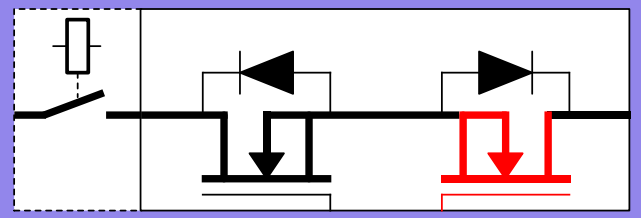
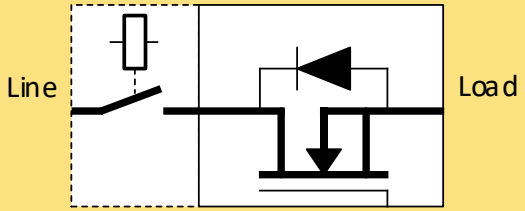
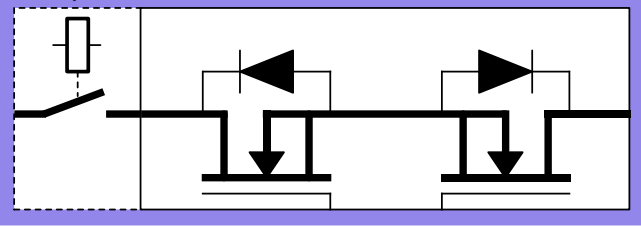
(expected < 1.2 cal/cm^2 by protection, source type, directionality to reduce required PPE for certain job tasks)

Source: [Current/OS System Reference]

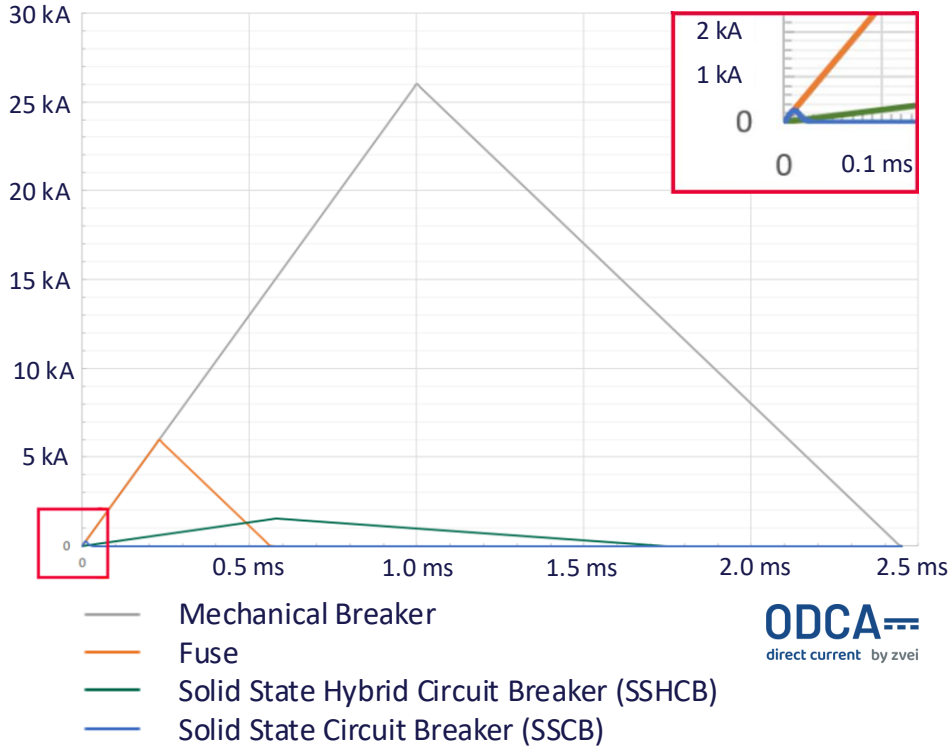


CurrentOS

Circuit Protection Types

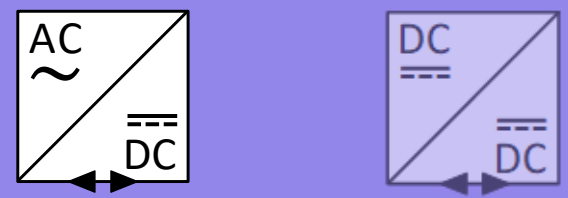
	Unidirectional Protection (e.g. Line and Load labeled)	Bidirectional Protection
Unidirectional Current Flow	• Prevents backfeed • May require external diode Example: 	• Operationally controlled to utilize one internal diode in conduction Example: 
Bidirectional Current Flow	• Reverse conduction often limited • Limited reverse protection only by isolation (few operations) Example: 	• Full functionality, higher cost Example: 

CurrentEDS



- **Directionality** and **protection** influence faults
- **SSCBs** simplify **selectivity**, greatly reduce fault energy
- **eFuses differ** from breakers **by excluding physical isolation**

Converter Types

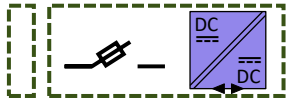
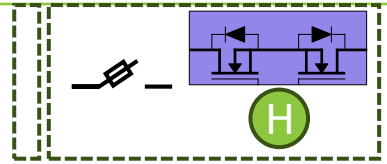
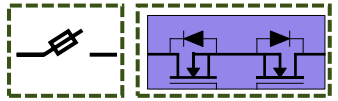
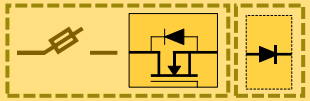
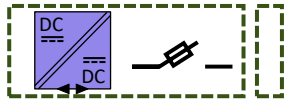
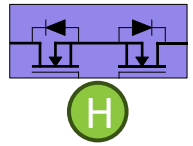
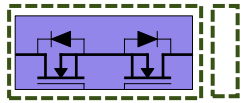
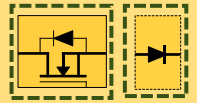
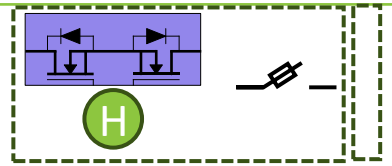
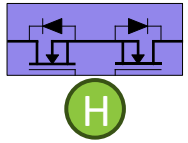
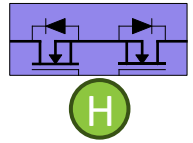
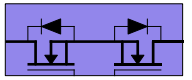
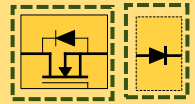
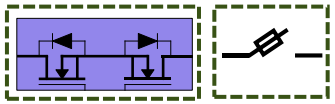
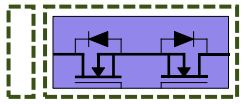
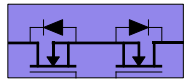
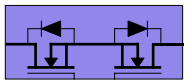
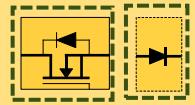
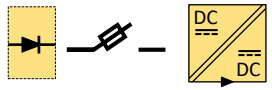
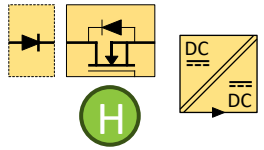
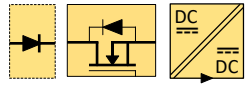
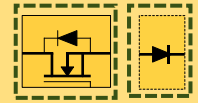
	Unidirectional Conversion	Bidirectional Conversion
Isolated	• Fault limiting, e.g. AC-DC iso unidirectional converter DC-DC iso unidirectional converter 	• Fault limiting, e.g. AC-DC iso bidirectional converter DC-DC iso bidirectional converter 
Non-Isolated	• Fault feeding, e.g. AC-DC non-iso unidirectional converter DC-DC non-iso unidirectional converter 	• Fault feeding, e.g. AC-DC non-iso bidirectional converter DC-DC non-iso bidirectional converter 

- **SSTs, TRUs** formed of multiple blocks
- **SSTs** are current limiting, can be unidirectional or bidirectional
- **TRUs** are current feeding OR limiting, can be unidirectional or bidirectional
- **SSTs** utilize **internal** MF/HF **xformer** for independent DC grounding
- **TRUs** utilize **external** LF **xformer** for independent DC grounding IF secondary ungrounded and dedicated

CurrentDS

Zone Transitions

Arrival in Zone 4 is by means of an SSCB and int. / ext. diode for directionality.

	To Zone 0	To Zone 1	To Zone 2.1	To Zone 2.2	To Zone 3	To Zone 4
From Zone 0	Direct		✖	✖	✖	✖
From Zone 1						
From Zone 2.1	✖					
From Zone 2.2	✖					
From Zone 3	✖					
From Zone 4	✖	✖				

H SSHCB

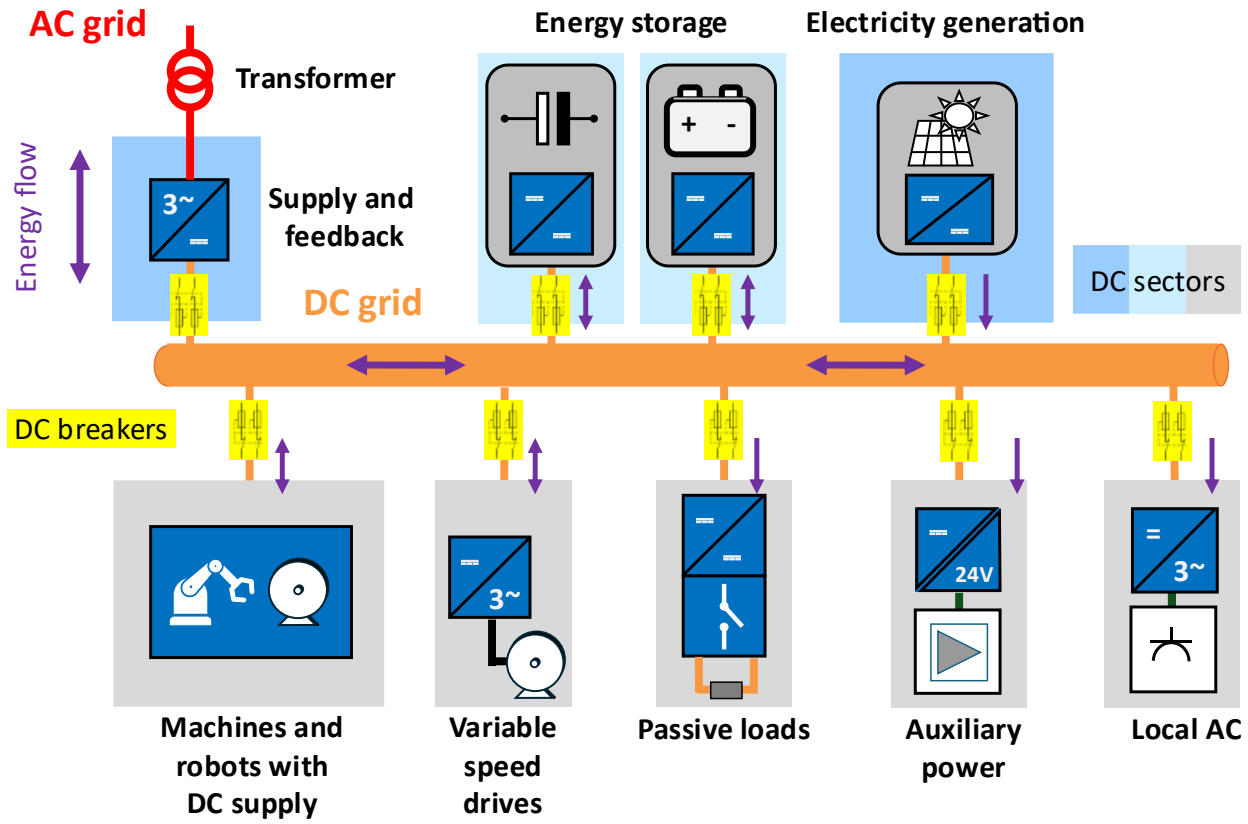
From | To

Converter types may vary dependent on unipolar / bipolar system

CurrentOS

DC Sectors: Overview

- Consist of **functionally related loads / sources**
- Connected to DC bus via **joint protection device**
- **Many parallel** sectors
- **Subsectors** possible
- Implemented and **industry-proven**



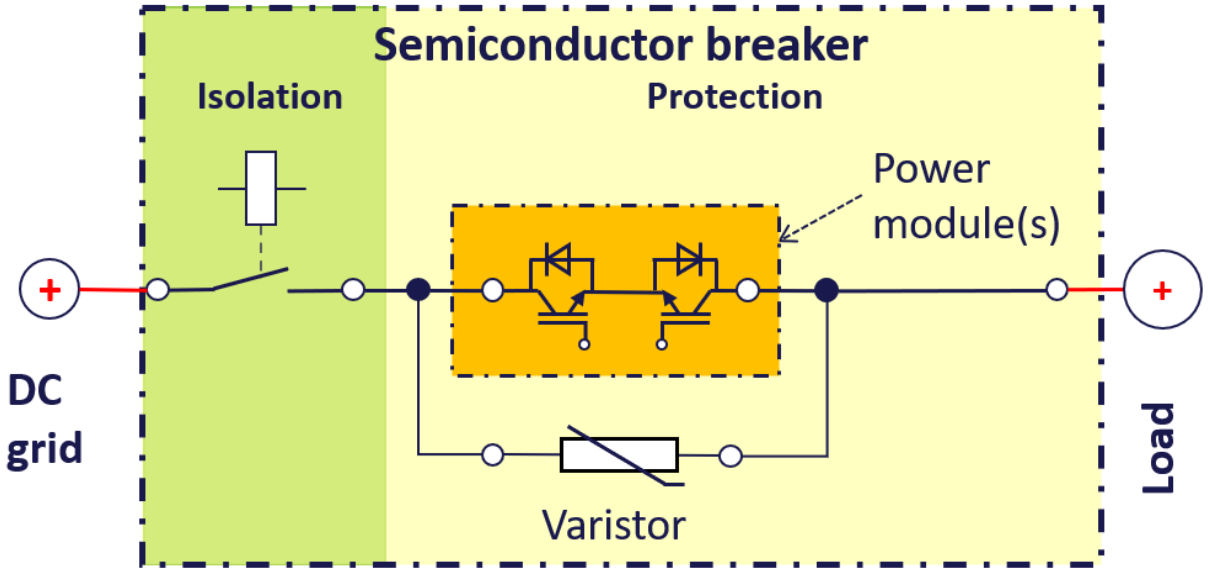
Source: [ODCA DC Microgrid Description]

ODCA direct current by zvei

DC Sectors: Semiconductor Breaker Protection

- **Selectivity → high availability**
  - Prevent fault propagation to grid or neighbors
  - Ensure continuity of service
  - Semiconductor breakers are the most reliable means for selectivity
- **Isolation → safe disconnection**
  - IEC 60947-…, UL 489I
- **Overcurrent** protection
  - µs time range
- **Pre-charging** function → prevent disconnection from in-rush currents
- **Monitoring** for current & voltage

Source: [ODCA DC Microgrid Description]



ODCA
direct current by zvei

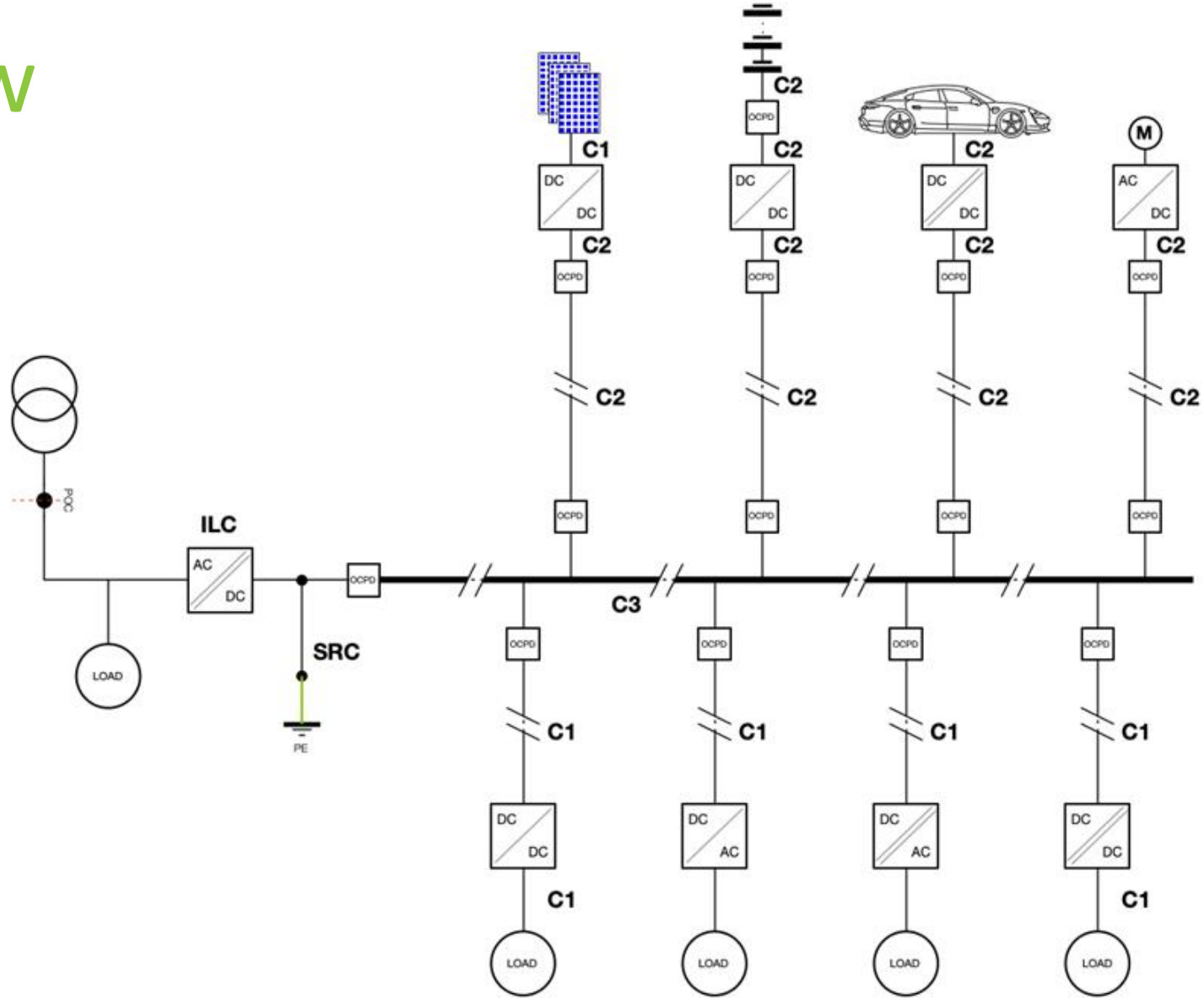
DC Circuits: Overview

IEC Definitions

- C1 = 1 source
- C2 = 2 sources (both ends)
- C3 = 3+ distributed sources
  - Protection devices with same tripping characteristic recommended in C3

SSCBs / SSHCBs recommended for low let-through energy

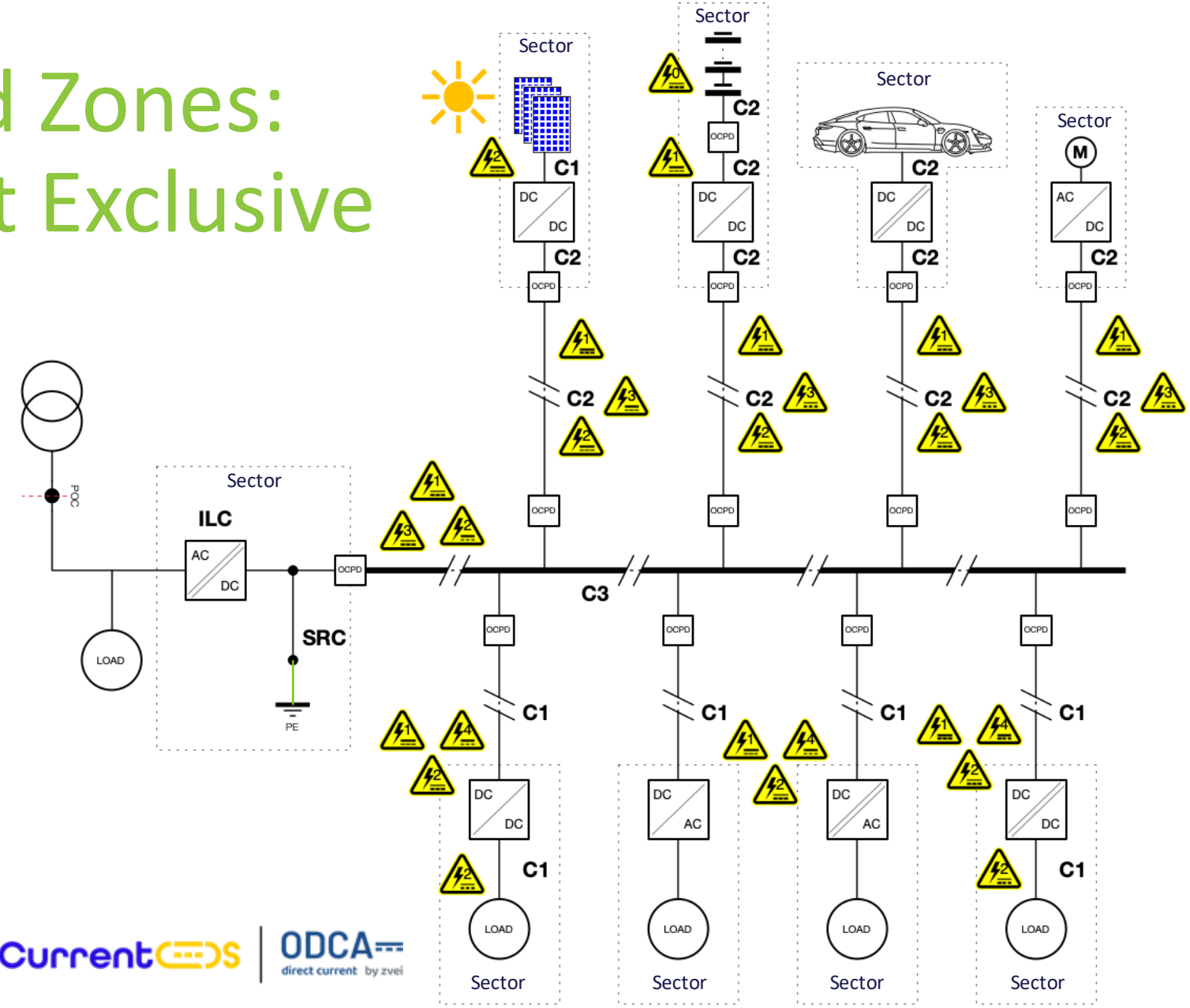
Source: IEC 60364-8-82 + AMD1



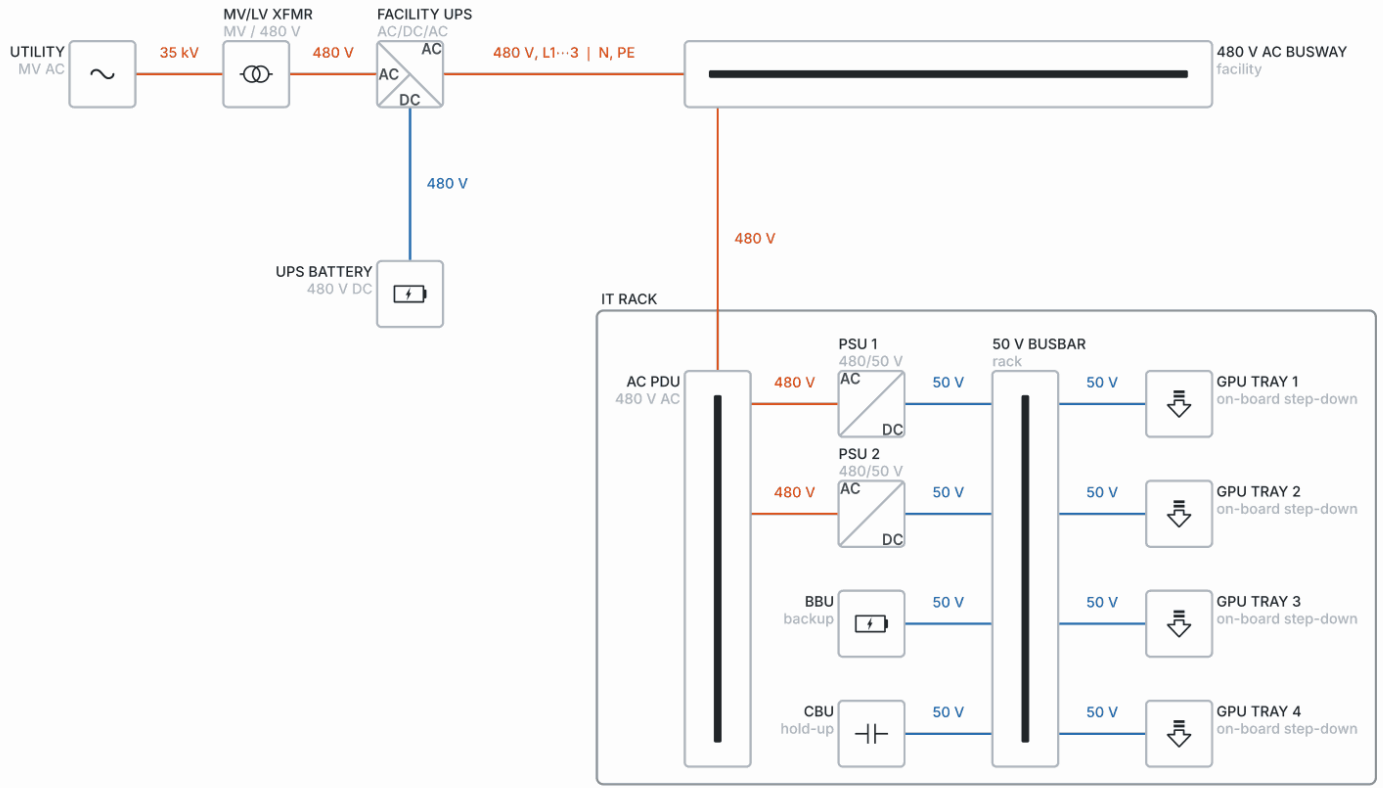
Circuits, Sectors, and Zones: Complementary, Not Exclusive

- Sectors may contain more than one type of circuit
- Sectors may contain more than one type of zone
- Circuits may correspond to different zones depending on source type, protection speed, and directionality

Key concepts from all three approaches are useful and complementary.



Same power, less copper and power conversions

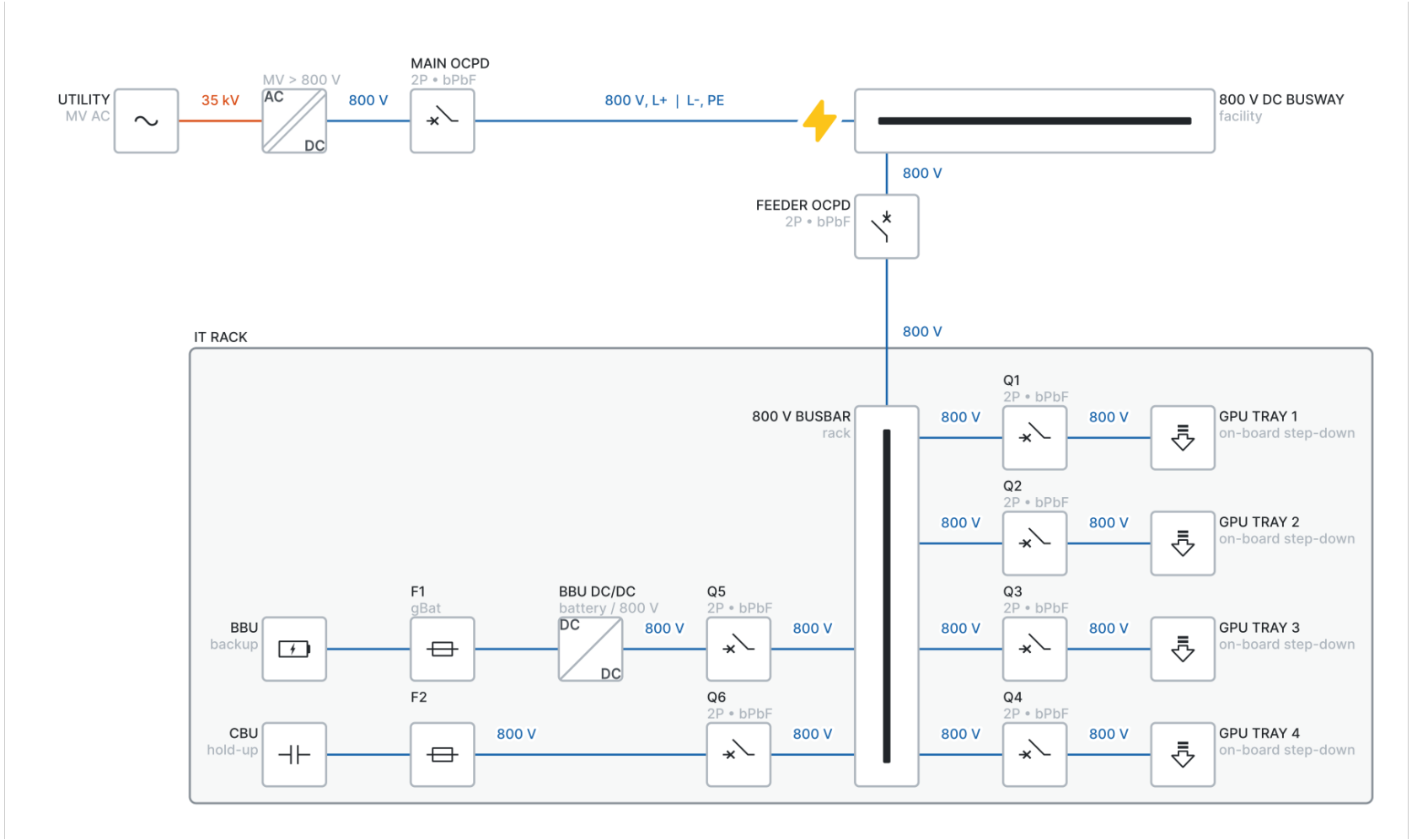


AC today, UPS in the room

A double-conversion UPS rectifies and inverts; the rack PSU rectifies again. Three stages before the tray.

4 conversion stages. 3 before the tray, 1 on the board.

One fault at the busway

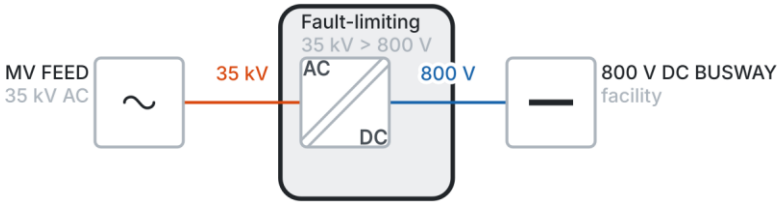


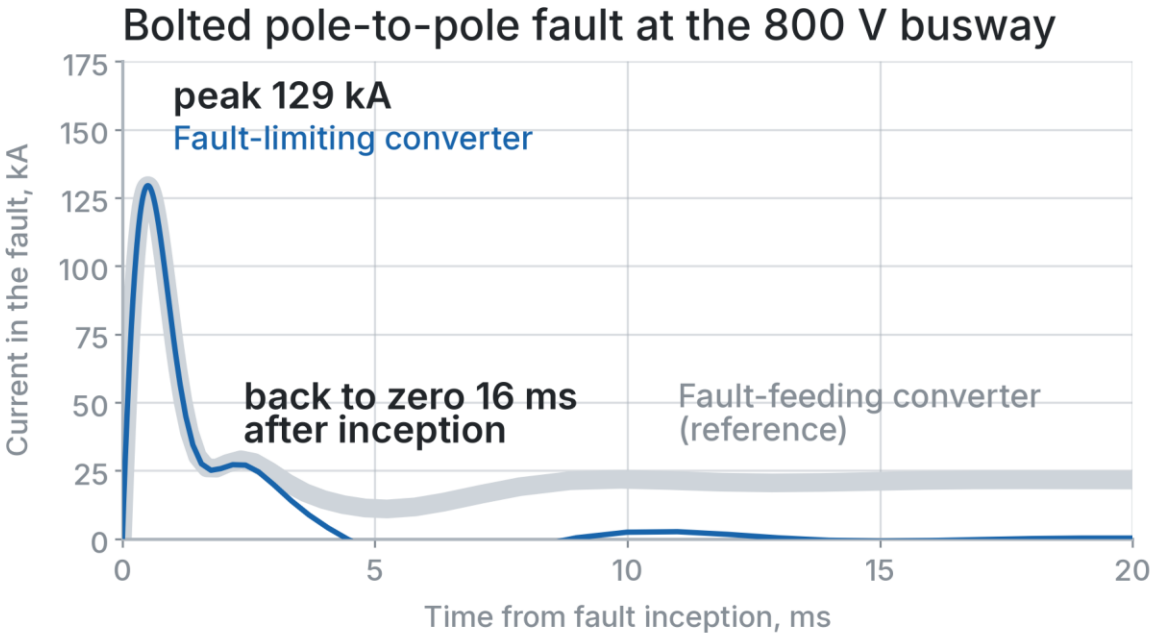
Example: AC and DC voltage may vary depending on utility, site architecture and readiness/availability of equipment

Same fault, same peak: one is fault-feeding, one is fault-limiting

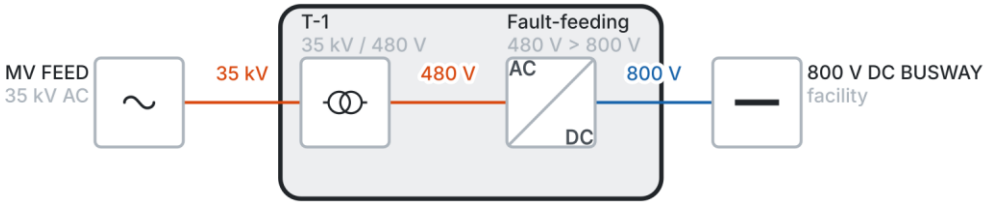
Simulated in DCIDE (EMTP model)

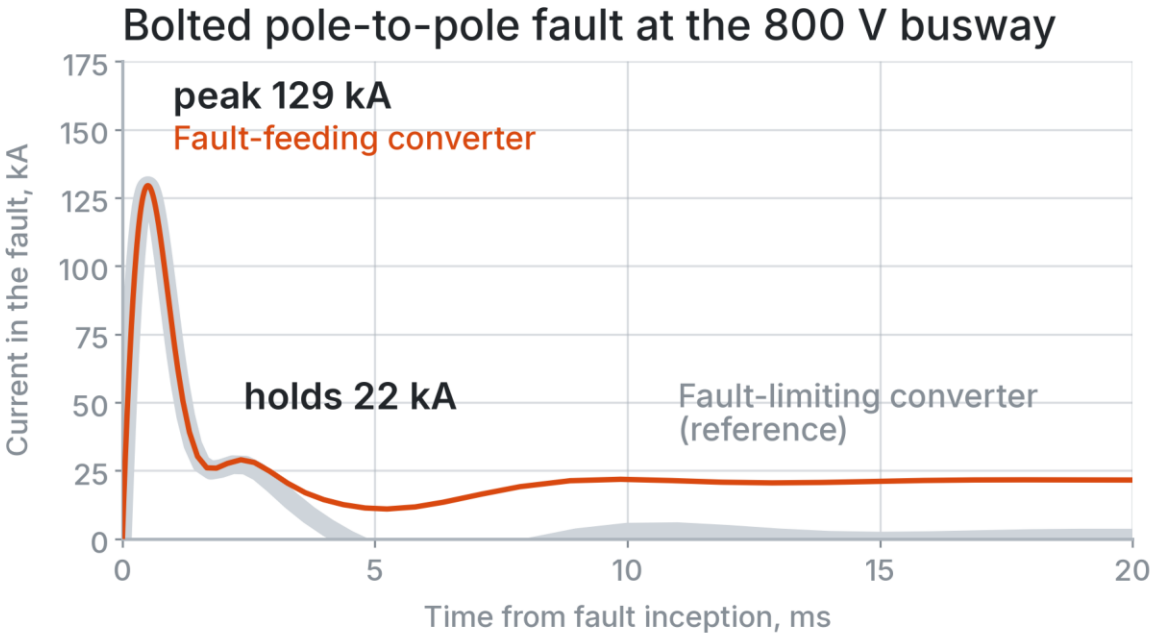
Fault-limiting example





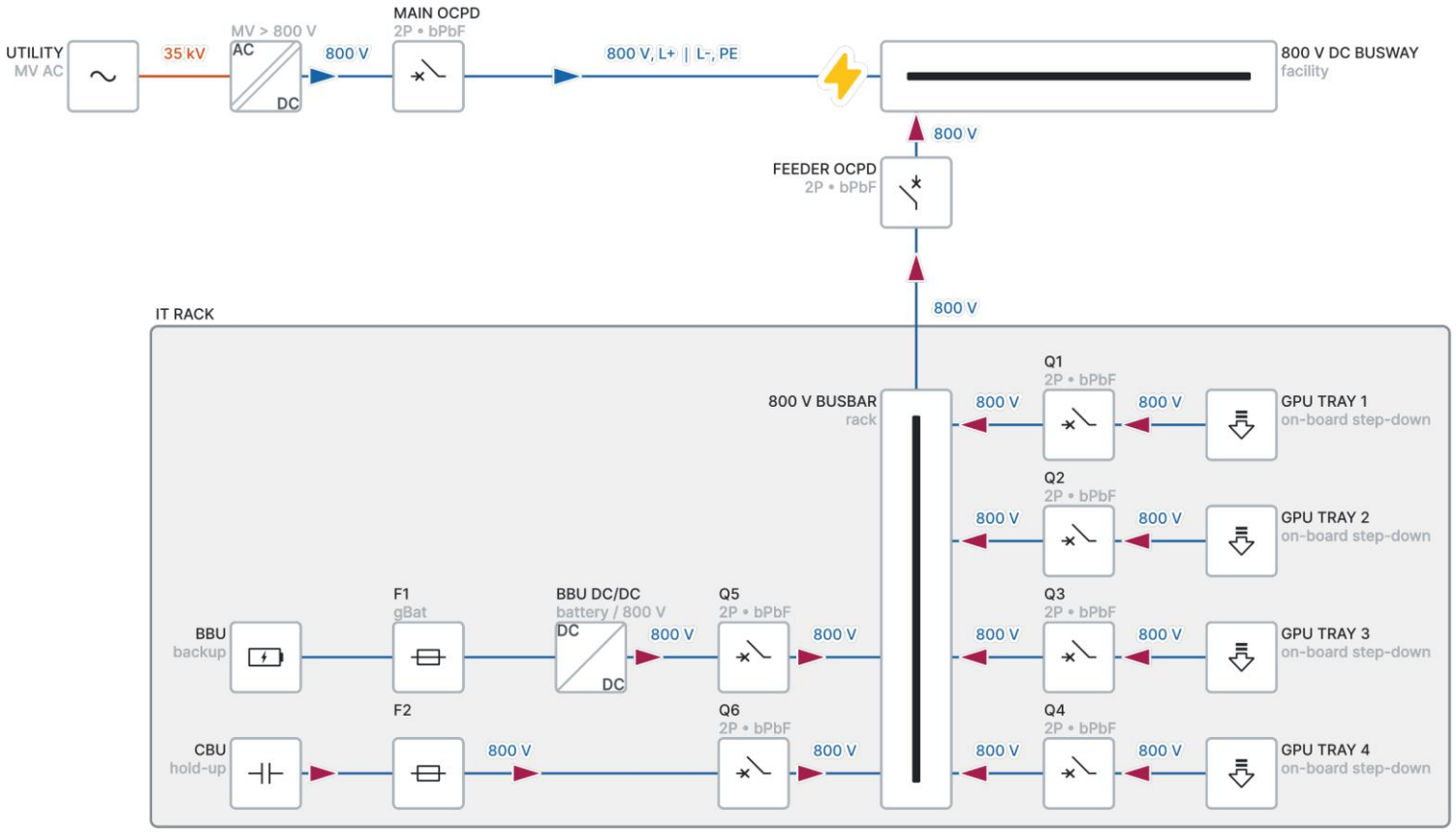
Fault-feeding example

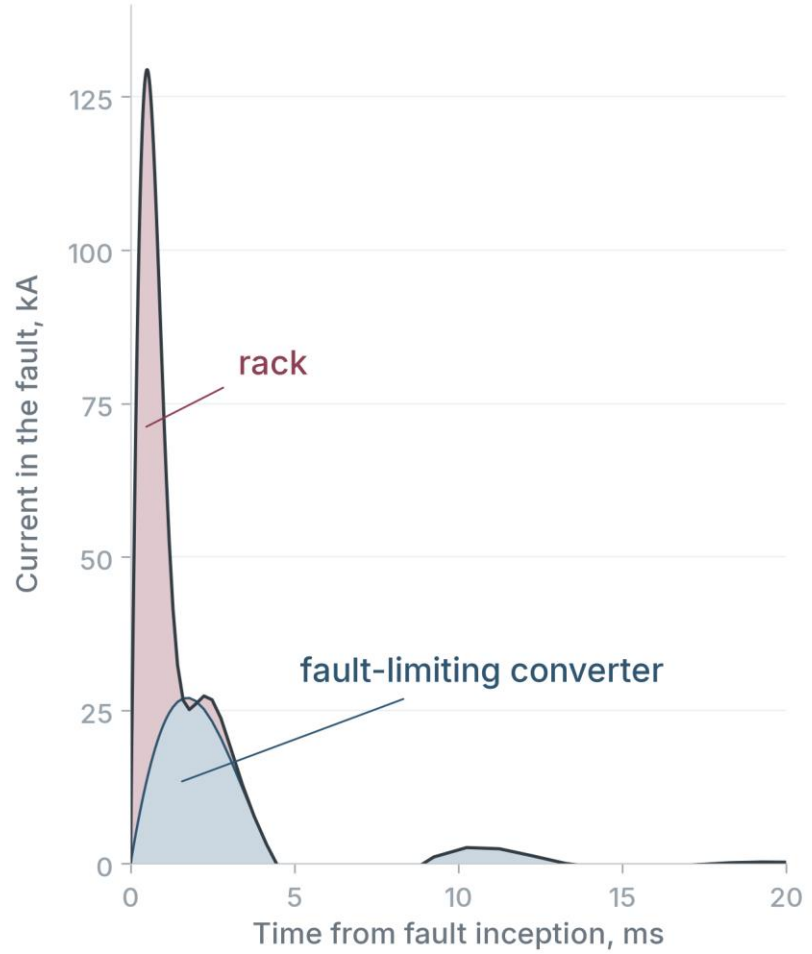




Prospective fault currents are exemplary and vary per case (technology dependent)

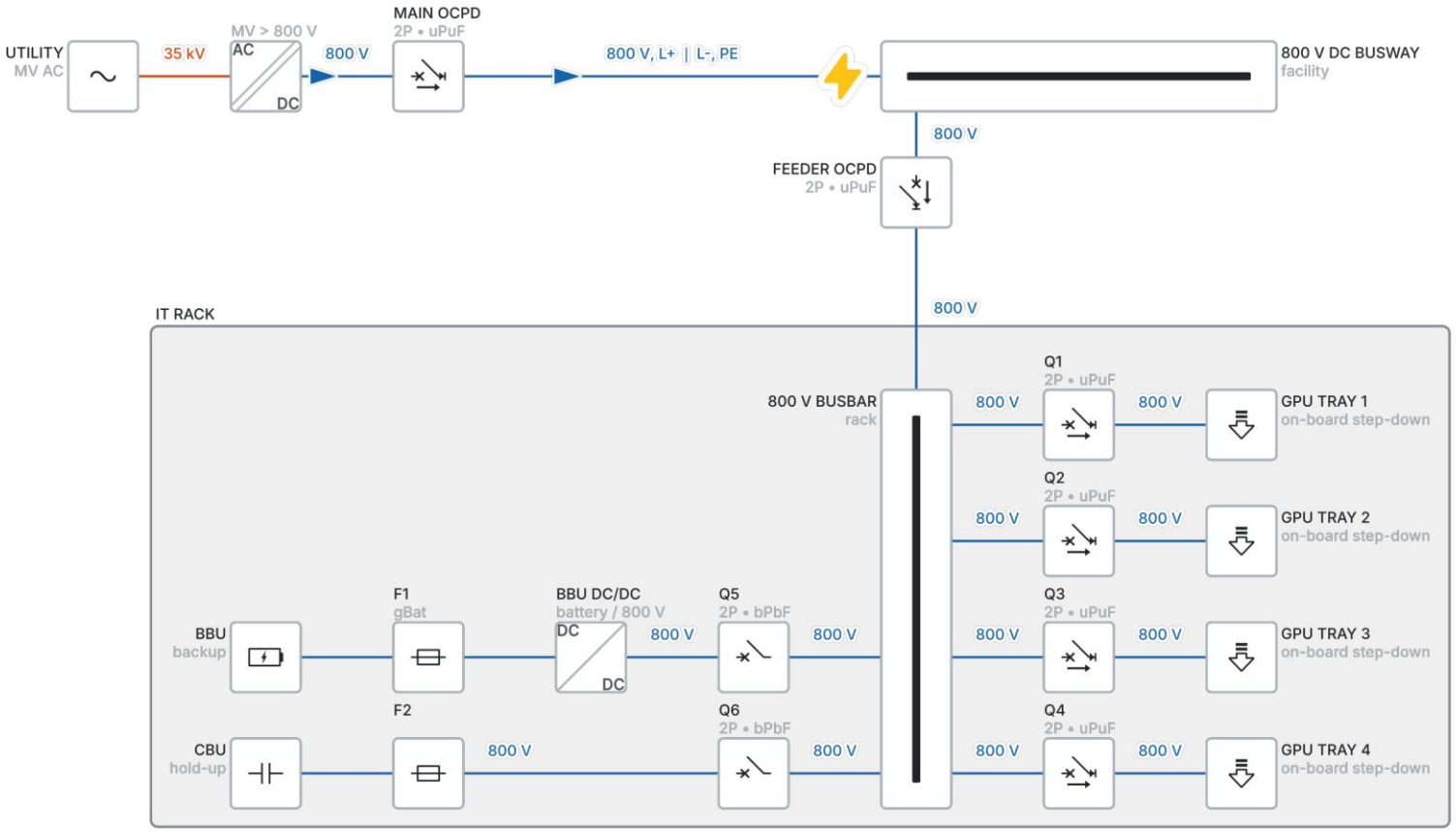
Default situation: rack and source contribute to fault current

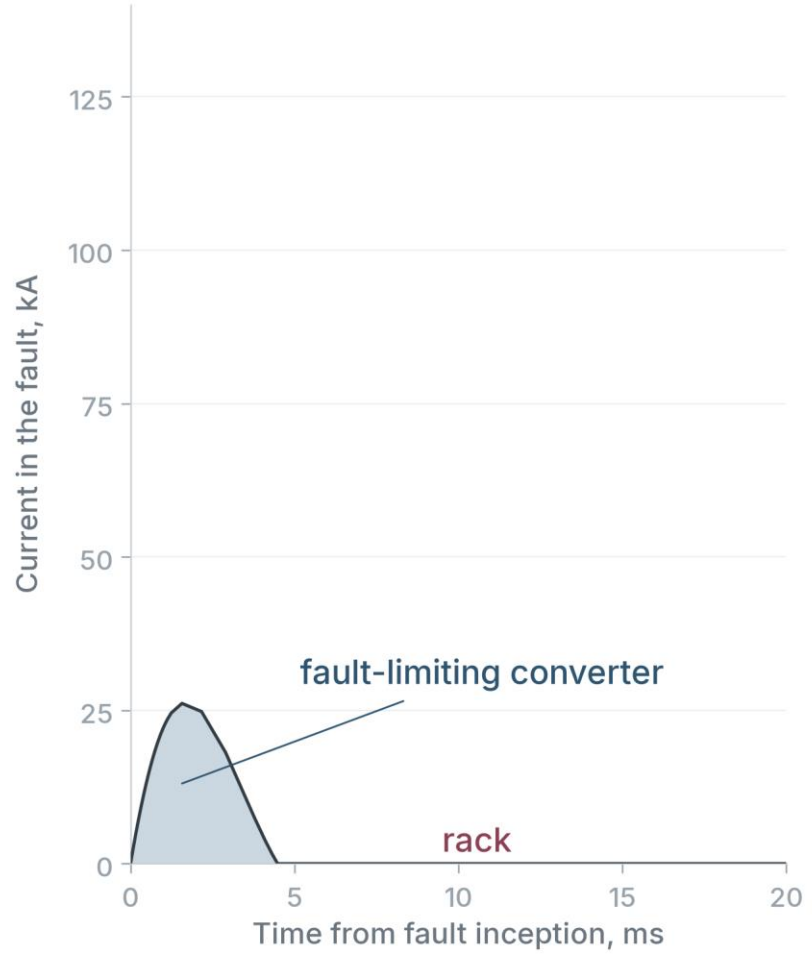




Interact with and simulate this architecture online:
https://dcide.app/projects/VnmJs:x5wyf:8E01g/editor

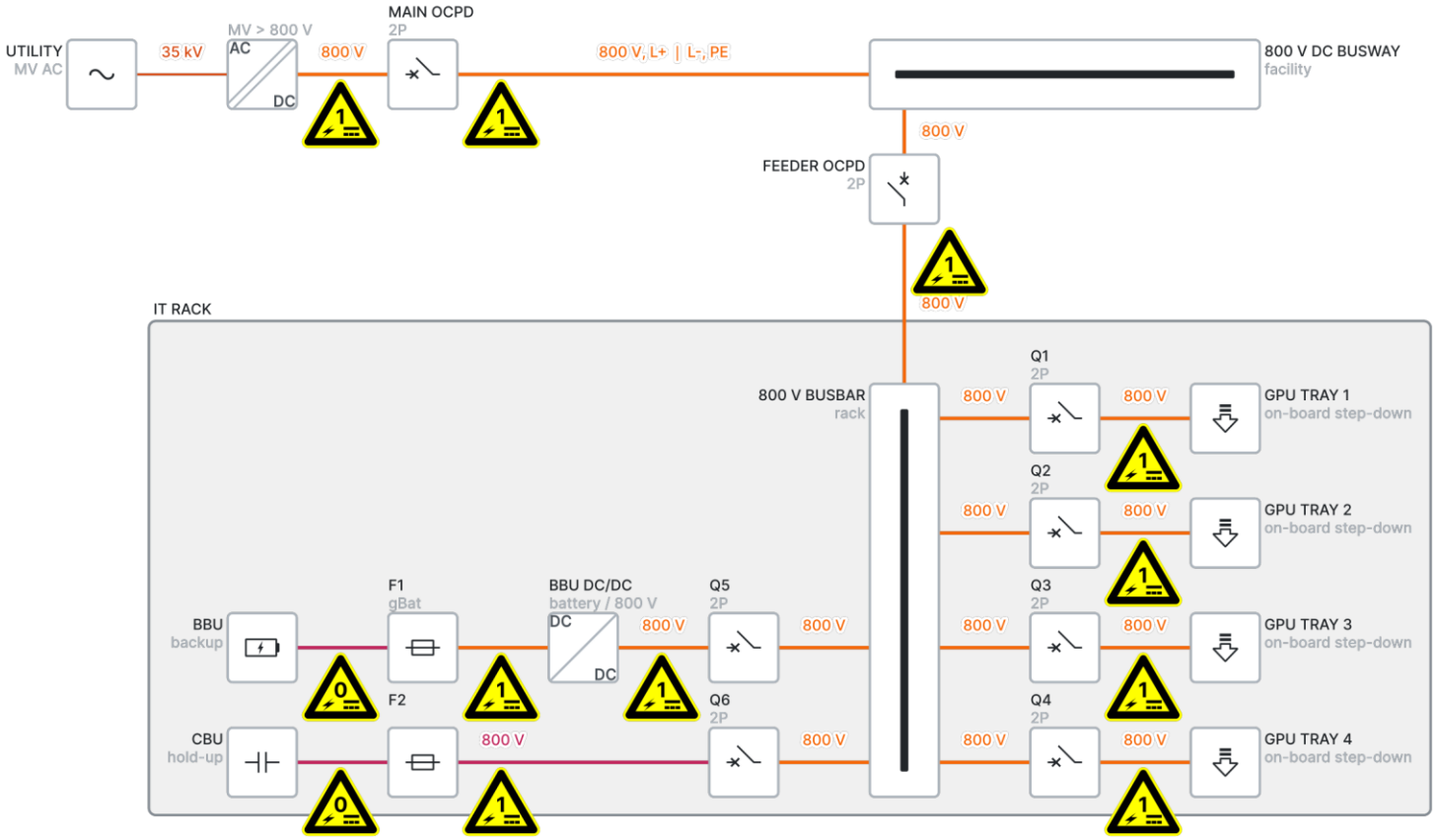
Unidirectional flow to avoid backfeed current



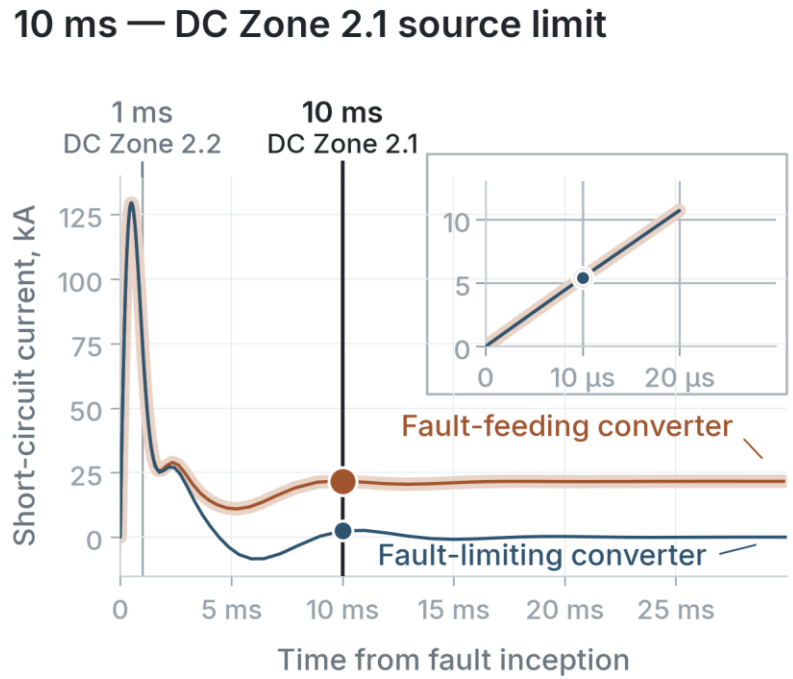


Interact with and simulate this architecture online:
https://dcide.app/projects/VnmJs:x5wyf:8E01g/editor

Conventional: DC Zone 1 and 2.1

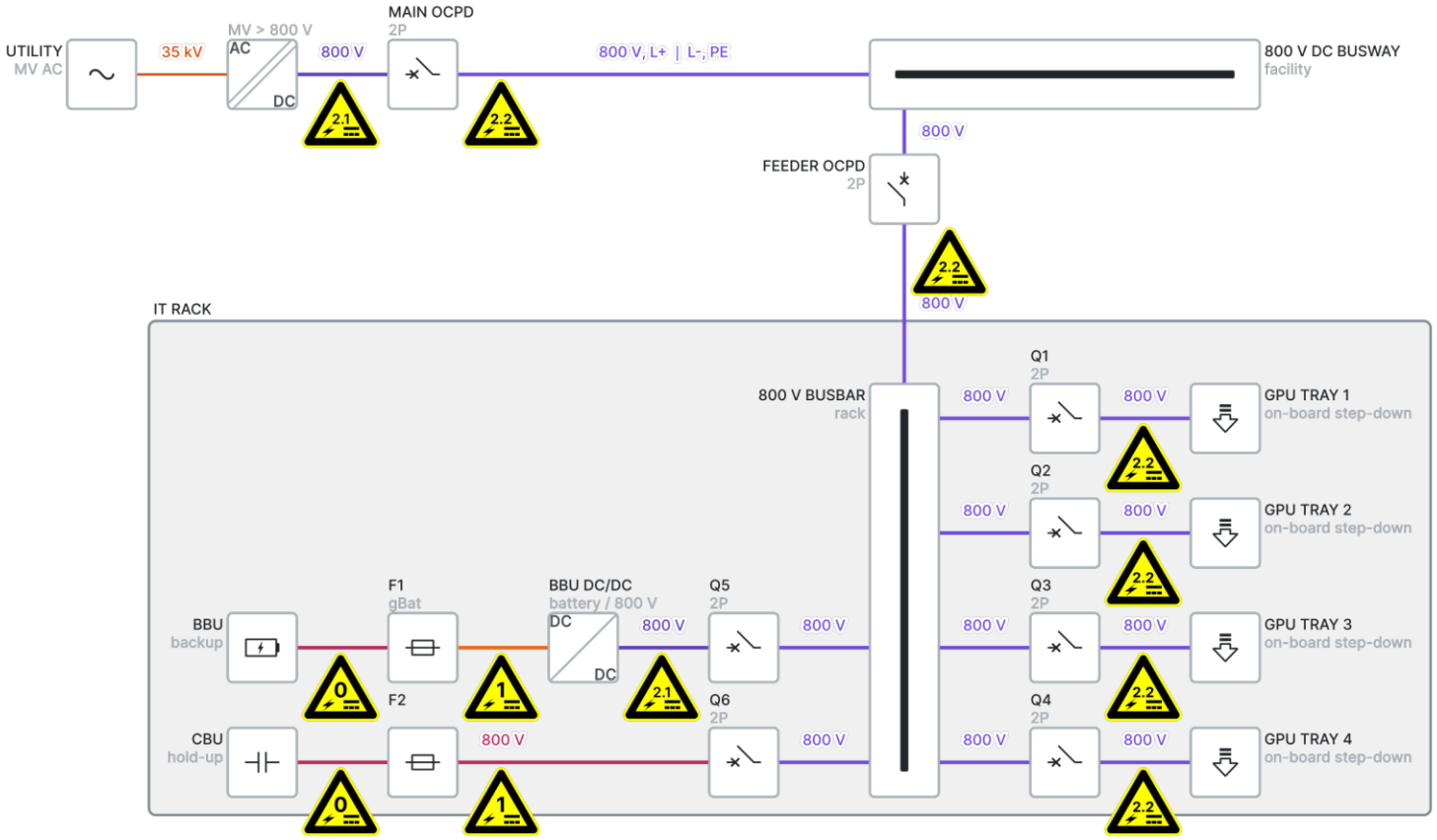


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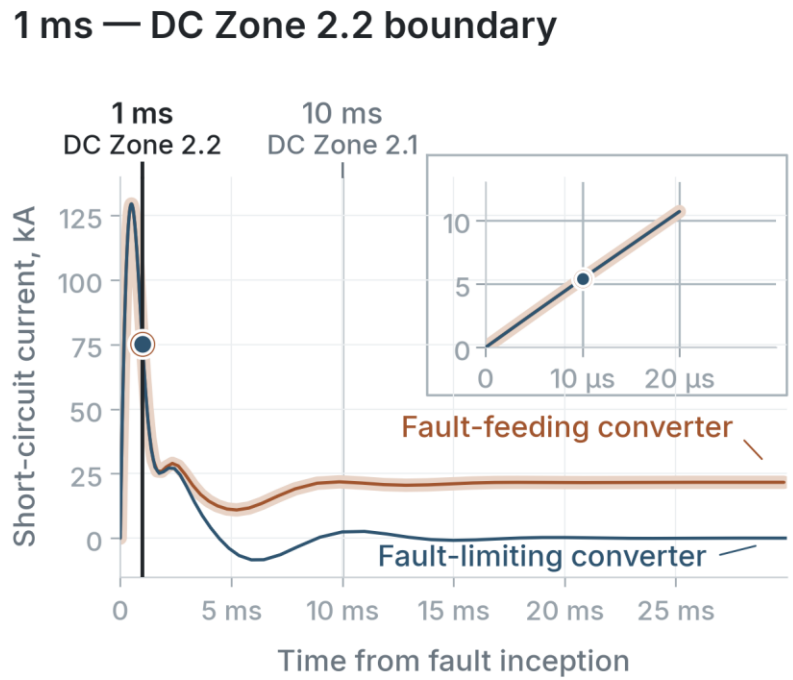


If two circuits with different zones are in the same space and unsegregated, the lower zone dominates for consideration of risk.

Hybrid breaker: DC Zone 2.2

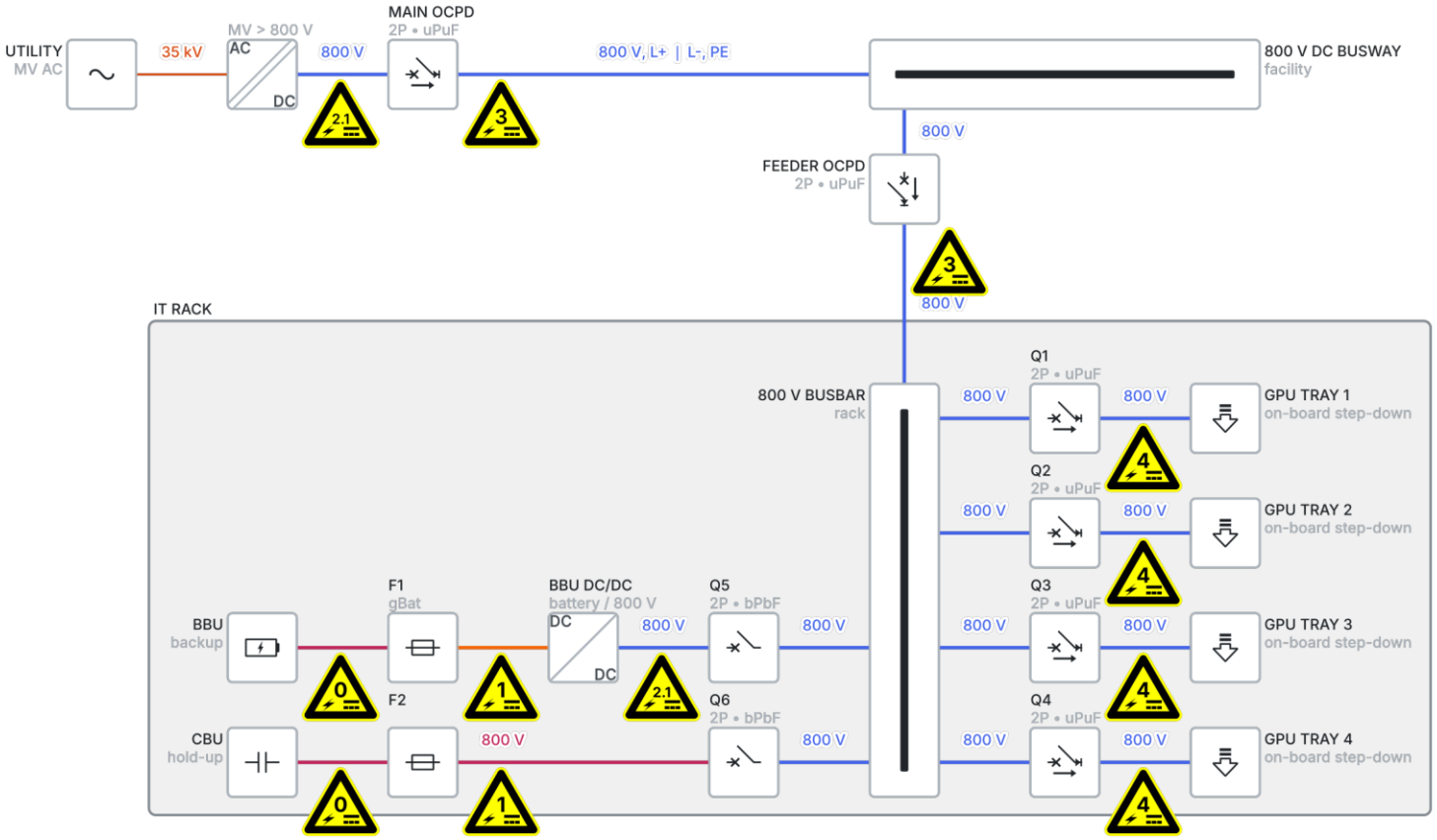


Interact with and simulate this architecture online:
https://dcide.app/projects/VnmJs:x5wyf:8E01g/editor

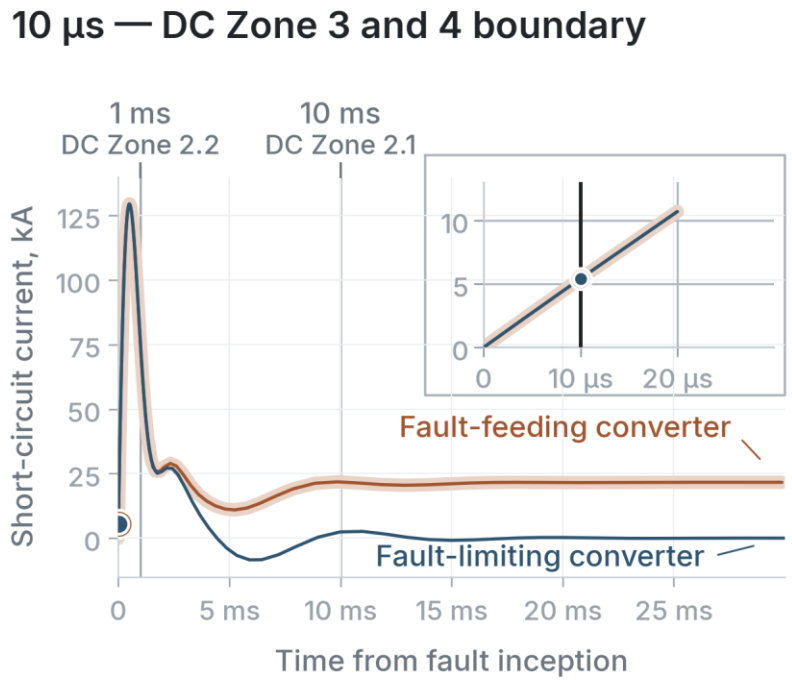


If two circuits with different zones are in the same space and unsegregated, the lower zone dominates for consideration of risk.

Solid-state breaker, uPuF: DC Zone 3 and 4



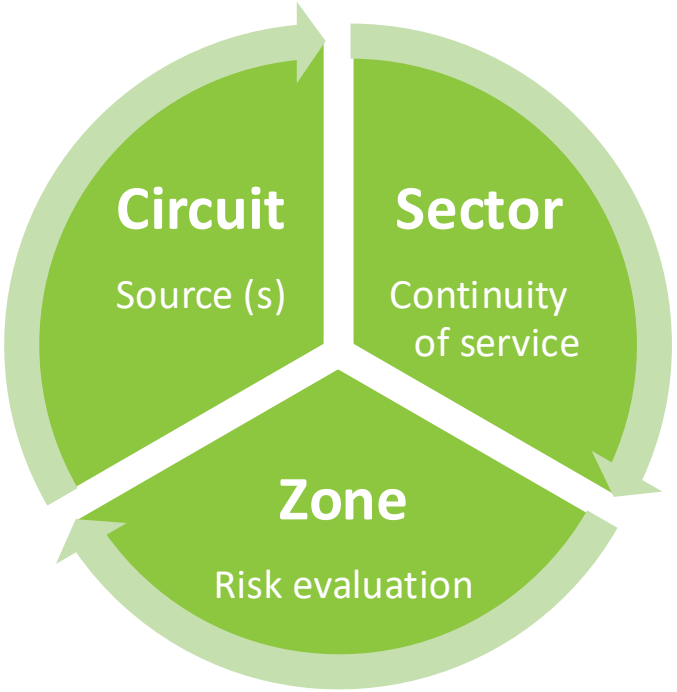
Interact with and simulate this architecture online:
https://dcide.app/projects/VnmJs:x5wyf:8E01g/editor



If two circuits with different zones are in the same space and unsegregated, the lower zone dominates for consideration of risk.

Call to Action

- Data Centers don’t have to reinvent Direct Current design
  - Sectors are a proven approach from ODCA to design-in selectivity for resilient operation
  - Zones are a proven approach from Current/OS to identify and proactively address risk (e.g. <1.2 cal/cm^2)
    - Alternate custom approaches to reduce incident energy could also be utilized
    - Zones are a natural consequence of source, protection, and directionality choices (easy visualization in tools like DCIDE)
  - Protection concepts for designing in low incident energy jointly explored by Current/OS, ODCA, and OCP



- How to get involved
  - Data Center Facility Power Distribution Sub-Project https://www.opencompute.org/community/power-distribution

Thank You!

